

Hei Shing Helson Go

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SUMMARY

Engineer, applied researcher, and startup co-founder with seven years turning ambiguous, high-stakes problems into validated, measurable results. Strengths in **quantitative modeling and optimization**, **leading cross-functional technical teams**, and **delivering complex systems under tight deadlines**, from first principles through to real-world deployment.

EXPERIENCE

Early-Stage Autonomous-Systems Venture

Oct 2025 – Present | Toronto, ON

Co-founder & Chief Engineer

- **Co-founded** a four-person early-stage venture and led its engineering, taking a control-and-navigation system from concept to a flight-tested prototype while exploring commercialization of the technology.

University of Toronto Autonomous Drone Racing Team

Sep 2024 – 2025 | Toronto, ON

Engineering Lead

- **Led** a 7-person team to build an autonomous racing system, placing **6th worldwide** at the 2025 Abu Dhabi Autonomous Racing League, setting technical direction to deliver against a fixed competition deadline.

Samsung AI Center

Jun 2019 – Aug 2019 | Montreal, QC

Research Intern

- **Translated** a novel machine-learning resource-allocation algorithm into a tested prototype and built a large-scale simulation to validate it.

AERO McGill

Sep 2016 – May 2019 | Montreal, QC

Technical Lead

- **Led** a 6-person student team to design and build an autonomous aerial system that won **2nd place** and the **Judges' Award for Professionalism** at the 2019 Unmanned Systems Canada national competition.

RESEARCH

University of Toronto Institute for Aerospace Studies

Sep 2020 – Oct 2025 | Toronto, ON

Ph.D. Researcher — Flight Systems & Control Lab

- **Developed and validated** optimization-based algorithms guaranteeing sub-0.5m positioning accuracy without GPS for collaborating autonomous vehicles, cutting positioning error by **over 75%** in simulation and **54–61%** on physical hardware; published across three peer-reviewed venues.

Industry collaborations: Applanix (Trimble) and Drone Delivery Canada

- **Built** a GPS-free LiDAR navigation system for Applanix (Trimble) and a data-driven control system for Drone Delivery Canada, each validated in live, real-world flight tests.

EDUCATION

University of Toronto

2019 – 2025 | Toronto, ON

Ph.D., Aerospace Science & Engineering

McGill University

2015 – 2019 | Montreal, QC

B.Eng., Mechanical Engineering

SKILLS

Quantitative & Analytical: optimization, statistical modeling, simulation, estimation and forecasting, data analysis, machine learning

Technical: Python, C++, Rust; rapid prototyping, test-driven development; published open-source control library, contributions to major engineering libraries

Leadership: cross-functional team leadership, technical direction, delivery under tight deadlines

PUBLICATIONS

Peer-reviewed publications in **IEEE ICRA** and the **AIAA Journal of Guidance, Control, and Dynamics**, among others (full list available on request).